MILITARY SPECIFICATION
REMOVER, AIRCRAFT CANOPY: M8A1
ASSEMBLY

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy: M8A1 Assembly, loaded with Cartridge, Aircraft Canopy, Remover, M150, for use in conjunction with other propellant actuated devices in aircrew emergency escape system.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F10522124	- Remover, Aircraft Canopy, M8A1 Assembly (conversion)
C8837890	- Carton, Packing, Ammunition for Remover, Aircraft Canopy, M8 with Cartridge M150
C9230161	- Box, Packing, Ammunition for (1) Remover, Aircraft Canopy, M8 with Cartridge M150
IEL-10522124	- Inspection Equipment List for Remover, Aircraft Canopy, M8A1 Assembly

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications and the following:

3.2 Torque.

3.2.1 Assembly.- An assembly torque of 500 plus 50 inch pounds (running plus bottoming torques) shall be applied between the head and adapter housing after insertion of cartridge.

3.2.2 Breakaway.- The head and adapter housing of the assembled remover shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.3 Functional.

3.3.1 Ballistics.- Removers loaded with accepted M150 Cartridge shall propel a 350 pound mass vertically and shall comply with the requirements listed in Table I.

TABLE I

Temperature	*Velocity, min. (fps)	Peak Thrust, Pounds Min.	Max.	Ignition Delay Max. (see 6.3)
-65° to -70°F	10.7	3500	7000	25 Milliseconds
65° to 75°F	10.7	3500	7000	25 Milliseconds
200° to 205°F	10.7	3500	9000	25 Milliseconds

* Velocity taken at tube operation.

3.3.2 Locked shut.- Removers loaded with accepted M150 Cartridges and temperature conditioned at minus 65 degrees F to minus 70 degrees F, and plus 200 degrees F to plus 205 degrees F, shall withstand the generated gas pressures with no evidence of leakage or mechanical failure (See 6.2) of exterior components when fired "locked shut" (See 6.4). The peak thrust shall not exceed 10,200 pounds over the temperature range.

3.3.3 The removers shall not fail to fire.

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3.3.4 The removers shall function without mechanical failure (see 6.2).

3.4 Workmanship.- The requirements for workmanship are as specified on the applicable drawing, referenced specification and the following:

3.4.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.4.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.4.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submittal of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

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4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 - 10000	200	2	3	3	4

4.2.2.1 Cartridge, Aircraft Canopy, M150, Drawing D10521960 covering a detail of Drawing F10522124.

	Defect	Method of Inspection
CRITICAL		
1.	Mixed cartridges	Visual
2.	Metal defective	Visual
3.	Sealing disc, missing, damaged or improperly secured	Visual
4.	Marking, missing, incorrect or illegible	Visual
MAJOR		
101.	Foreign matter	Visual
MINOR		
201.	Case scratch, wrinkled, bulged or dented	Visual
202.	Varnish, missing or inadequate	Visual

4.2.2.2 Assembly Dwg. F10522124 (before assembly of head).

CRITICAL		
1.	Gas port hole obstructed	Visual
2.	Cartridge missing	Visual
3.	Firing pin missing	Visual

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4.2.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly, Drawing F10522124.

CRITICAL	
1. Mixed removers	Visual
2. Marking, missing, incorrect or illegible	Visual
MAJOR	
101. Protective finish, inadequate or defective	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
201. Shipping plug missing	Visual
202. Location of labels	Visual

4.2.2.4 Packing in Inner Box, Drawing C8837890.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Top of box, loose	Manual
203. Strapping, missing, loose or improperly assembled	Manual & Visual
204. Supports or fillers, missing from box	Visual

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4.2.2.4(Cont'd)

Defect	Method of Inspection
MINOR	
205. Box damaged	Visual
206. Seal, missing or improperly positioned	Visual
207. Motor wires loose	Visual
208. Marking, missing, incorrect or illegible	Visual

4.2.2.5 Packing in Outer Box, Drawing C9230161.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual
202. Liner, fillers, cushions or spacers, missing	Visual
203. Strapping, missing, loose or improperly assembled	Visual
204. Hardware, missing, loose or improperly engaged	Visual
205. Box damaged	Visual
206. Contents of box exposed	Visual
207. Car seal, missing or improperly sealed	Visual
208. Marking, missing, incorrect or illegible	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- Each head and adapter housing of the assembled remover shall be subjected to the breakaway torque application. Failure of any remover to comply with the requirement of 3.2.2 shall be cause for rejection of the remover.

4.2.3.2 Functional tests.- The following tests shall be performed by the Government.

4.2.3.2.1 Ballistics.- Eighteen units shall be tested as specified in 4.3.2 to determine compliance with the requirements of 3.3.1, 3.3.3, 3.3.4 and Table I. The units shall be temperature conditioned as provided in Table III and 4.3.2.3. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

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TABLE III

Temperature	Sample Size
-65° to -70°F	6
65° to 75°F	6
200° to 205°F	6

4.2.3.2.2 Locked shut.- One remover temperature conditioned at minus 65 degrees F. to minus 70 degrees F. and 2 removers at 200 degrees F. to 205 degrees F. shall be fired "locked shut" (See 6.4). The units shall be temperature conditioned as specified herein and 4.3.2.3. Failure of any unit to comply with 3.3.2 and 3.3.3 shall be cause for rejection of the lot.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10522124.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench set at the torque requirement specified in 3.2.2 for the head and adapter housing shall be used to determine the adequacy of the joint. The assembly shall be secured in a fixture and the breakaway torque applied to each head and adapter housing.

4.3.2 Functional tests.

4.3.2.1 Ballistics.- The remover samples shall be rigidly supported in the fixture by the mounting points. The remover tube shall be secured to a carriage and allowed to move freely while propelling the weight specified in 3.3.1 when test fired. The remover shall be initiated by means of an M3A1 Initiator or simulated pressure with sufficient force to activate the firing pin. The remover shall be tested within 10 minutes after removal from the conditioning chamber. Samples shall be reconditioned at the original temperature if this time is exceeded.

4.3.2.2 Locked shut.- The remover samples shall be rigidly supported in a fixture and initiated by means of an M3A1 or simulated pressure with sufficient force to activate the firing pin. The remover shall be tested within 10 minutes after removal from the conditioning chamber. Samples shall be reconditioned at the original temperature if this time is exceeded.

4.3.2.3 Temperature conditioning.- Prior to conducting the ballistic test and locked shut test specified in 4.3.2.1 and 4.3.2.2 respectively, the assembly samples shall be subjected to the temperature conditions specified in 3.3.1 and 3.3.2. A simulated assembly loaded with a dummy cartridge containing a temperature sensitive element shall be placed in the temperature conditioning chamber

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4.3.2.3 (Cont'd)

at the same time as the remover samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature. Upon completion of the required temperature conditioning, each remover shall be tested as specified herein.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- (Level A) - Packing and marking shall be in accordance with Drawings C8837890 and C9230161. Marking shall be changed to specify Remover, Aircraft Canopy, M8A1.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 "Ignition Delay" is defined as the time that elapses between the firing of the cartridge primer and the beginning of the continuous rise in the thrust time record.

6.4 "Locked shut" is defined as a condition in which the inside tube is restrained from moving upon firing the remover.

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